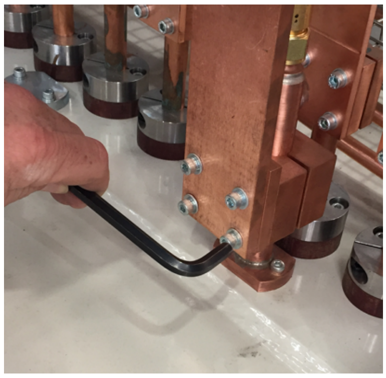
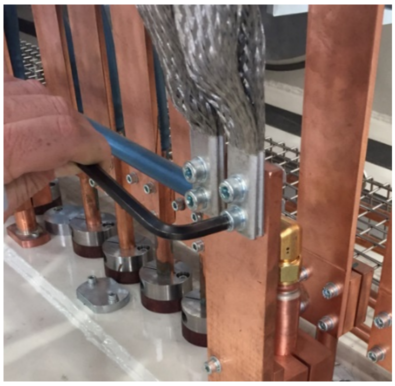
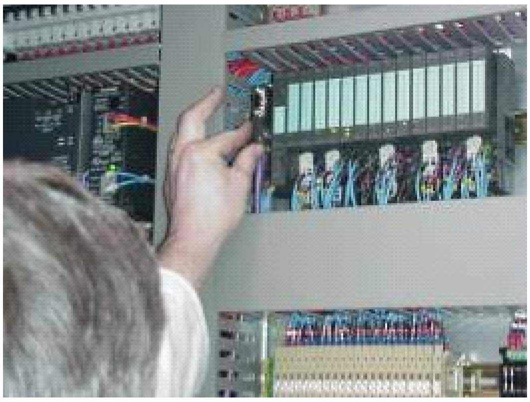
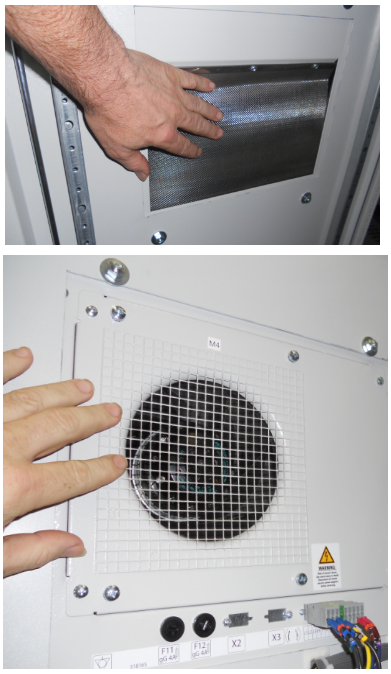
Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Check all bus bar connections to stop potential overheating at joints
Figure 12.2-209 Example of a technician checking all bus bar connections to stop potential overheating at joints	
Quarterly	Check the condition of the connections of the internal busbar system. Tighten bolts As required.
Figure 12.2-210 Example of a technician checking the internal busbar system	
Figure 12.2-211 Example of a technician checking the internal busbar system	

Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	Check that all the connections on between busbars are tight. Tighten bolts as required.

Maintenance Requirement	Equipment & Task to Be Performed
5 Years	Change the o-ring seals and insulators on the leadthroughs

12.2.31 Electrical Panels & Air Conditioner Units

	**DANGER**
	Before commencing with any maintenance and/or cleaning work on the panels disconnect the electrical power supply.

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check that all the communication cable terminations are fully inserted
Figure 12.2-212 Example of a technician checking the communication cable terminations	
Monthly	With the electrical panels isolated, use a vacuum cleaner to remove dust etc. Do not allow build up.
Monthly	Check the condition of any forced air filters
Monthly	Check the operation of the door switches do switch off the air conditioners and bring up a alarm on the HMI
Monthly	Wipe down the inside of the unit to remove any dust build up
Figure 12.2-213 Example of a technician cleaning the internal fan of the unit	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Wipe down the heat exchanger fins after removing the access panels.
Figure 12.2-214 Example of the cleaning unit's heat exchanger with the external panels removed	
Monthly	Wipe down the external access panels.
Figure 12.2-215 External access panels	
Monthly	Visually check the unit to panel interface to ensure that no leakage paths have occurred.

DANGER

Before commencing with any maintenance and/or cleaning work on the panels disconnect the electrical power supply.

Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	Check tightness of all connections.
Bi-Annual	Check that all interconnecting cables between plc input/output racks are inserted correctly.

Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	Check the condition of all the contactors used.
Bi-Annual	Check the operation of all isolator switches
Bi-Annual	If available, use an infrared image camera to check all connections in the panels.

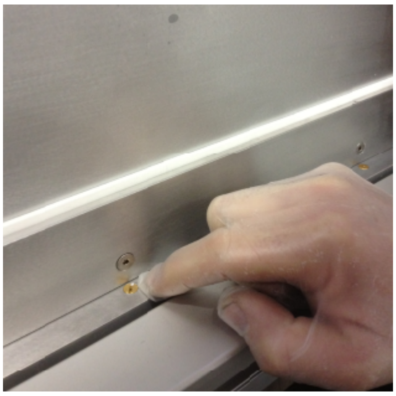
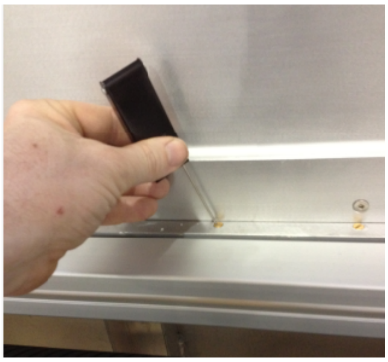
DANGER

Before commencing with any maintenance and/or cleaning work on the panels disconnect the electrical power supply.

Maintenance Requirement	Equipment & Task to Be Performed
3 Years	Replace UPS battery as per manufacturer’s guidelines.

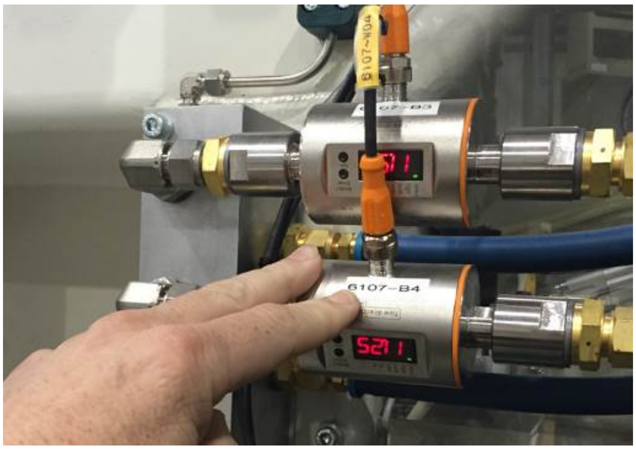
12.2.32 Planar Plasma Treater

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check insulating materials for indication of over heating
Figure 12.2-216 Example of a technician visually checking for damage to the insulating materials on a plasma treater	
Monthly	Check tightness of all fixing components.
Monthly	Check for wear on plates by physical erosion.
Figure 12.2-217 Example of a technician visually checking for physical erosion of a plasma treater	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check gas distribution system for even and constant gas flow. Disconnect internal gas distribution line and blow compressed air through the distribution system.
Figure 12.2-218 Example of a technician visually checking for any blockage of the gas wedge nozzles	
Figure 12.2-219 Example of a technician identifying the gas distribution connection to be used to check for gas distribution	
Figure 12.2-220 Example of a technician using “welding torch” cleaners to clean a blockage in a gas wedge nozzle	

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Visually check for damage to the high voltage cables.

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Check the connections for tightness and damage, internally and externally
Figure 12.2-221 Plasma supply internal connection	
Figure 12.2-222 Plasma supply external connection	

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Record the water flow passing through the treater on the HMI screen or IFM flow sensors. Record and compare to ensure that there is no reduction over a period of time.
Figure 12.2-223 Example of a technician checking the water flow feedback from the sensors in the return legs of the water supply	 

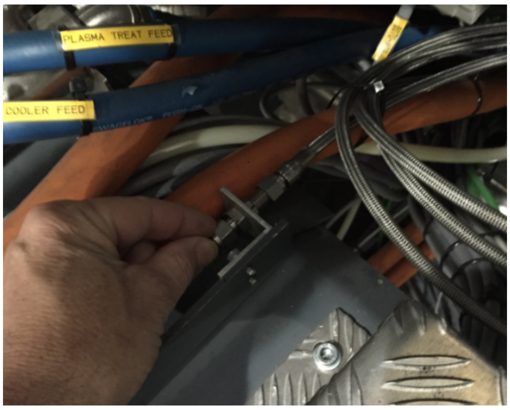
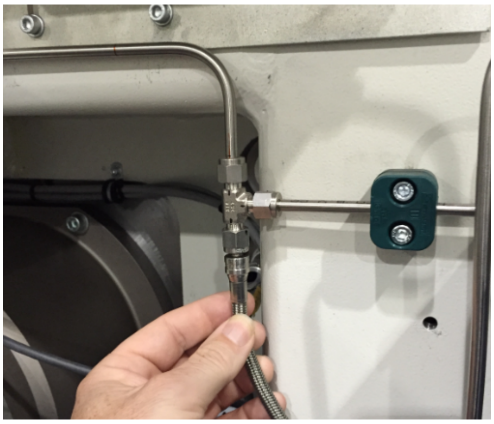
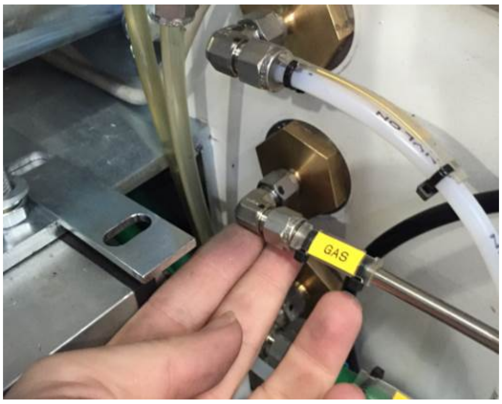
Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	Check the tightness of the High Voltage cable leadthrough connections
Figure 12.2-224 Example of a technician checking the tightness of the high voltage leadthrough connections	

Maintenance Requirement	Equipment & Task to Be Performed
Annual	Replace o-rings, where necessary

Maintenance Requirement	Equipment & Task to Be Performed
2 Years	Schedule BML field engineer to carry out rebuild/inspection of the treater including magnetic field checking

12.2.33 Plasma Treater – Gas Delivery System

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Visually check the gas bottle regulators daily to ensure that the delivered pressure does not exceed 2 bar (30 Psi). Pressures above this level can cause damaged to mass flow controllers.

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check the physical condition of flexible gas distribution hoses and replace as necessary.
Figure 12.2-225 Example of a technician checking the condition of the external gas lines	
Figure 12.2-226 Example of a technician checking the condition of the external gas lines	
Figure 12.2-227 Example of a technician checking the condition of the internal gas lines	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Use industry recommended practices to check for gas leaks from the lines and fittings. A guideline is BML document 327G821 The reason for such a regular procedure, is that escaping gas can cause a risk to the users

Snoop® liquid leak detectors detect gas leaks in hard-to-reach areas. Sustained bubble action works even on very small leaks and vertical surfaces. Flexible snooper tube extends for hard-to-reach areas. Does not contain chlorine, aliphatic amines, or ammonium compounds. Dries clean, without staining.

	Important!
	Bobst Manchester Ltd, recommend the use of the specialized leak detection fluid "Swagelok Snoop® Liquid Leak Detector". As it complies with the following standards • **MIL-PRF-25567 Leak Detector Compound, Oxygen Systems, Type I, 1 to 70°C (33 to 158°F)** • **NFPA 52 Section 6-12.2 Leak Testing Compressed Natural Gas Vehicular Fuel System** • **EPA Part 60, Appendix A, Method 21, Section 4.3.3 Alternative Screening Procedures** https://www.swagelok.com/en/product/Leak-Detectors-Lubricants-Sealants/Snoop-Liquid-Leak-Detector

	DANGER!
	All necessary precautions should be adhered with when handling compressed air systems. **Good working practices should be maintained in regards to oxygen enriched environments**

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check the gas supply regulators etc., to ensure that they are still within their manufactures life cycle.
Figure 12.2-228 Example of a technician checking the gas delivery bottle regulator	

Cleaning & Assembly Procedure For Oxygen Use Components

This document describes the requirements for the cleaning, degreasing & assembly of component parts intended for use in oxygen-enriched environments.

Preparation

Pre-cleaning of parts should be performed before final cleaning for oxygen service to remove any gross contamination such as large quantities of oils, greases, and inorganic particulates.

All cleaning & pre-assembly must be done on a properly cleaned stainless steel work surface as far away from sources of contamination as possible. All assembly tools and assembly surfaces should be cleaned with 99% Isopropyl alcohol prior to use. Care should be taken not to touch part surfaces that will be in direct oxygen service except with clean, powder-free, nitrile gloves.

Cleaning Procedure

Parts are to be cleaned/degreased using a proprietary cleaning agent 99% Isopropyl alcohol.

Take all necessary precautions. Read the MSDS sheet.

After degreasing dry parts completely by purging with nitrogen.

Note: - system compressed air should not be used for drying parts

Visually inspect parts for the presence of contaminants under strong white light and for the absence of accumulations of lint fibres. Any visual contaminate is cause for re-cleaning.

Cleaning Materials

Solvents/Detergents: - 99% Isopropyl alcohol.

Cleaning cloths should be clean, lint free & free from oil or grease. Cleaning cloths should be removed after use.

Protective Equipment:-

Disposable non-powdered nitrile or latex gloves to be worn when handling cleaned parts.

Suitable PPE to be worn (determined by cleaning agent).

Assembly Procedure

All assembly tools and assembly surfaces should be cleaned with 99% Isopropyl alcohol prior to use. Care should be taken not to touch part surfaces that will be in direct oxygen service except with clean, powder-free, Nitrile gloves (Latex gloves are organic and therefore should not be used when assembling oxygen equipment).

Note: - PTFE tape contains oil-based hydrocarbons & must never be used for thread sealing anywhere on the oxygen supply line.

Only thread sealing products suitable for oxygen use should be used (Cobas Green Oxygen Thread Sealing Tape – Part No 268166G).

Parts should be assembled immediately after cleaning.

Pre-assembly must be done on a properly cleaned stainless steel work surface as far away from sources of contamination as possible.

Assemble all parts onto the machine.

Care must be taken to avoid contamination during assembly.

Any item which becomes contaminated during assembly must be re-cleaned before installation.

When assembly is complete purge each Oxygen system with Nitrogen.

Leak test each system with Nitrogen.

A suitable oxygen rated leak test solution is required for the leak checking e.g. “Snoop” by Swagelok.

Do not use a soap based leak test solution as it will contain fats and oils

Gas Line Identification

All gas lines to be clearly labelled at connection points and at 2-metre intervals along the pipeline.

Labels must identify the specific gas type (oxygen, nitrogen, argon, etc.)

Further References

EIGA – IGC Doc. 33/06/E – Cleaning of equipment for oxygen services

12.2.34 Machine Safety Equipment

	Note !
	It is strongly suggested that records are kept of the testing of the all the safety circuits and equipment, for future reference

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Check the activation of the emergency stop circuit removes power from all rotating elements on the machine.
Daily	Check the operation of the G-stop circuit isolator removes power from rotating components when the traverse isolator is switched to the Off position
Daily	Check that when the switch in the “ON position” the audible siren sounds intermittently to warn all users that the machine is in a “UNSAFE” condition.
Figure 12.2-229 The location of the traverse lockout isolator	
Daily	Check the operation of the audible alarm, when the winding cart or drive system is in operation.
Daily	Visually check that the warning labels, as the machine was delivered are in place. If not replace ASAP

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Check the operation of the visual flashing orange beacons when the winding cart or drive system is in operation.
Figure 12.2-230 The visual beacon	

	Note !
	It is strongly suggested that records are kept of the testing of the all the safety circuits and equipment, for future reference

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check all local electrical isolators operate correctly (if installed)

12.2.35 Machine Water Services

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Check for leaks on all water manifolds and interconnecting rubber hoses, and make any necessary repairs.

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check for correct pressure differential across the water system manifolds. Check the values on the HMI water diagnostic screen and compare with previous records, to understand if there has been any major changes in the supply system, etc.
Weekly	Check all functions of pressure sensors on all manifolds. Close feed and return valves to the circuits and monitor the HMI diagnostic screen for a water shortage in these conditions.
Figure 12.2-231 Check all functions of pressure monitoring equipment	

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check that all sensor cables are connected correctly, and there is no damage to the wiring or devices.
Figure 12.2-232 Check that all sensor cables are connected correctly.	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Water treatment chemical analysis carried out to minimise potential corrosion

	**Warning !** **It is strongly recommended that water treatment of open and closed loop systems has to be carried out to ensure that the equipment does not fail**

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Open drain valve on the main filter unit to bleed off any entrapped debris.

Maintenance Requirement	Equipment & Task to Be Performed
Bi-annually	Remove and clean all in-line filter. This may need to be carried out more regularly, depending upon the water quality used.
Figure 12.2-233 Remove and clean in-line filter